JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING (AFFILIATED TO JNTU-GV)

DEPARTMENT OF MECHANICAL ENGINEERING

★ OFFICIAL ACADEMIC SYLLABUS & BLUEPRINT ★

B.Tech Mechanical Engineering R23 Engineering Mechanics Syllabus & Blueprint

Course Code:	R23-MEC-1102	**Academic Scheme:**	I B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Unit I: Statics of Particles & Rigid Bodies

Forces, Vector operations, Equilibrium of particles in 2D and 3D, Free body diagrams, Moment of force about a point and axis, Couples, Varignon's Theorem, Equivalent force systems.

Unit II: Friction & Centroid / Center of Gravity

Laws of dry friction, Angle of friction, Wedge friction, Ladder friction, Centroid of composite areas and curves, Pappus-Guldinus theorems, Center of gravity of composite bodies.

Unit III: Area Moment of Inertia & Mass Moment of Inertia

Parallel and perpendicular axis theorems, Polar moment of inertia, Radius of gyration, Moment of inertia of standard and composite shapes, Mass moment of inertia of slender rods, plates, and cylinders.

Unit IV: Kinematics of Particles & Rigid Bodies

Rectilinear and curvilinear motion in Cartesian and normal-tangential coordinates, Projectile motion, Work-energy principle, Impulse-momentum theorem, Impact of elastic bodies.

Unit V: Dynamics & Simple Mechanical Vibrations

D'Alembert's principle, Equations of motion for rigid bodies in plane motion, Introduction to free and damped harmonic vibrations, Natural frequency, Torsional vibrations.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Continuous Internal Evaluation (CIE)	30 Marks (2 Mid Exams + Assignments + Quizzes)
Semester End Examination (SEE)	70 Marks (Part A: 10 Short Questions = 20M, Part B: 5 Units choice = 50M)
Total Assessment Credits	3 Credits (3-0-0 Lectures/Tutorial/Practical)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM